

LDA and PCA

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Index Terms—Linear Discriminant Analysis, Principal Component Analysis, dimensionality reduction.

I. LDA/PCA

A. Questions

a) Question 1: According to the paper, what are the main differences between LDA and PCA?

LDA is supervised: it needs categorical class labels (a dependent variable), and finds a projection direction that separates the classes. Its goal is to maximize between-class scatter over within-class scatter, and the resulting axes represent linear discriminants.

The paper does not address two relevant differences that explain a key difference in its result. First, LDA caps at $C - 1$ components. Second, LDA is invariant to feature rescaling: scaling the data set will only introduce the same scaling factor in the between-class scatter matrix and the within-class scatter matrix, and since LDA is about finding a vector that maximizes the Rayleigh quotient, where the numerator has the between-class scatter and the denominator has the within-class scatter, this introduced scale factor appears in both numerator and denominator and the result doesn't change.

PCA is unsupervised: it ignores the class labels (the dependent variable), instead it only finds principal components (directions) that maximize variance in the data set, and the resulting axes represent principal components (orthogonal directions of maximum variance).

In addition to the paper's stated differences: PCA has no limit on components. It is also scale-sensitive on account of focusing on variance, which scales with the square of the measurement units; this is why standardization gave a smaller average percent error on PCA in the paper.

b) Question 2: Which proposal was better in terms of average percent error?

LDA, with a 14.57% average error, versus 17.83% for PCA with preprocessing and 19.56% for PCA without preprocessing.

c) Question 3: Numerically, what is the impact of dimensionality reduction techniques?

PCA without preprocessing reduces the average percent error by 2.33 percentage points (10.6%), with preprocessing it reduces the error by 4.06 percentage points (18.5%), LDA reduces the error by 7.32 percentage points (33.4%).

Notice that this baseline only uses two out of 215 attributes; this analysis measures reduction against weak feature

selection, not against full dimensional regression.

d) Question 4: Explain in your own words what the proposal LDA-PLS is, as per your understanding. Try to think on possible improvements or future exploration work.

LDA-PLS is a proposed improvement of LDA that uses the partial least squares method on the features matrix $\mathbf{X}$, but swapping the dependent variables vector $\mathbf{y}$ with $\mathbf{X}$ projected in the direction of LDA's calculated projection $\mathbf{w}_{LDA}$. Then the projection of $\mathbf{X}$ in LDA's calculated direction is subtracted from $\mathbf{X}$, the result of PLS is concatenated into a results matrix $\mathbf{B}_{LDA-PLS}$, the projection calculated from LDA is concatenated into an LDA results matrix $\mathbf{W}_{LDA-PLS}$, and the process continues until we have gone through all the latent variables. **ex-LDA-PLS** is a weighted sum of the outputs of plain LDA and LDA-PLS via a λ factor, where $\lambda = 0$ only leaves the result of LDA, $\lambda = 1$ leaves only the result of LDA-PLS, and anything in between blends both results. PLS-LDA uses PLS to reduce the data and then runs LDA on it; whereas LDA-PLS gets first a direction from LDA and rotates it with PLS.

Future work can be done into extending LDA-PLS into multi-class variables (right now everything is binary; this is a non-trivial extension), a better way to find λ than simple brute force, evaluating the computational cost of LDA-PLS (the authors didn't), or implementing kernelization for non-linear boundaries.

e) Question 5: What are the benefits of the proposal compared to traditional methods?

The benefit is that ex-LDA-PLS beats traditional methods on accuracy score in large datasets, whereas LDA-PLS beats traditional methods on small datasets; ex-LDA-PLS seems to overfit in smaller data sets. ex-LDA-PLS has an accuracy score of 0.9922 vs LDA 0.9576 on the Gas dataset; PCA collapses to 0.4958 on the Raman dataset; LDA-PLS reaches full accuracy with under 5 components where PCA needs about 15. LDA-PLS fixes LDA's "blind spot" without abandoning it, by making the direction of the projection adjustable.